Exploratory Data Analysis on the Iris Dataset

A Data Science Internship Report

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Tools: Python, Pandas, Matplotlib, Seaborn

1. Introduction

Exploratory Data Analysis (EDA) is a critical first step in any data science workflow. It enables an analyst to understand the underlying structure of a dataset, detect anomalies, test assumptions and form hypotheses prior to modeling. In this report we apply a complete EDA pipeline to the well-known Iris dataset, originally introduced by British statistician and biologist Ronald A. Fisher in 1936.

2. Objective

The objective of this project is to perform a thorough EDA on the Iris dataset in order to:
- Understand the distribution of each feature.
- Quantify the relationships between features.
- Identify which features best discriminate between species.
- Detect outliers, duplicates and missing values.
- Communicate the findings via clear, professional visualizations.

3. Dataset Description

The dataset contains 149 samples evenly distributed across three Iris species: setosa, versicolor and virginica (50 samples each). Each sample is described by four numeric features measured in centimeters: sepal length, sepal width, petal length and petal width.

Descriptive statistics

	sepal_length	sepal_width	petal_length	petal_width
count	149.00	149.00	149.00	149.00
mean	5.84	3.06	3.75	1.19
std	0.83	0.44	1.77	0.76
min	4.30	2.00	1.00	0.10
25%	5.10	2.80	1.60	0.30
50%	5.80	3.00	4.30	1.30
75%	6.40	3.30	5.10	1.80
max	7.90	4.40	6.90	2.50

4. Methodology

The analysis follows a standard EDA pipeline implemented in Python:

1. Load the CSV using Pandas.
2. Inspect shape, dtypes, missing values and duplicates.
3. Clean - remove duplicates, coerce dtypes, drop NaNs.
4. Compute descriptive statistics and class balance.
5. Compute pairwise correlations.
6. Detect outliers with the IQR rule.
7. Visualize distributions and relationships.
8. Summarize per-species behavior.

5. Data Cleaning

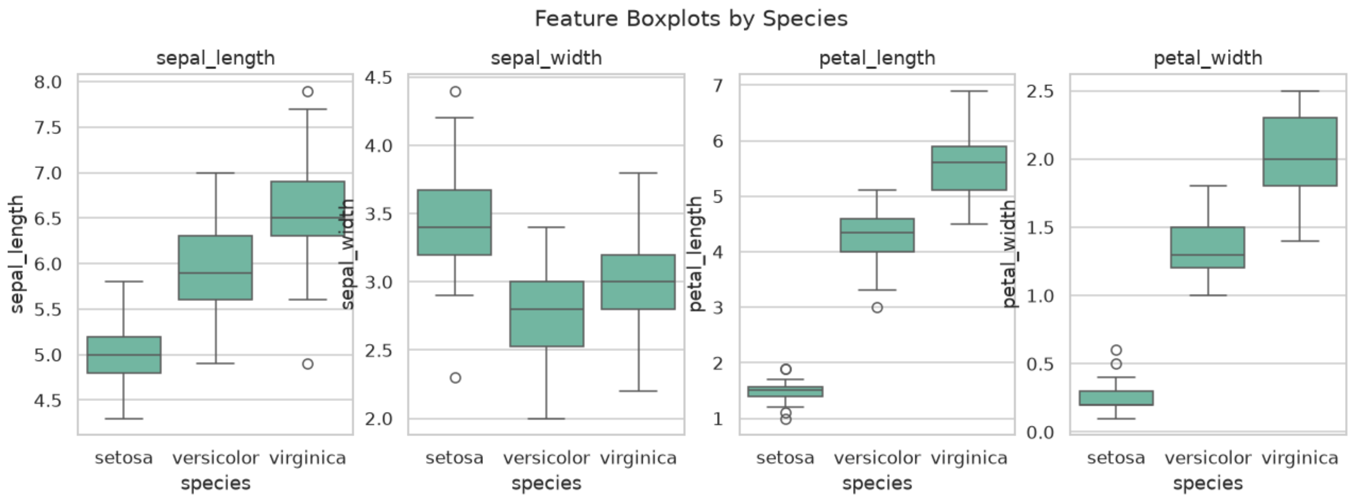
Missing values: 0. Duplicate rows after deduplication: 0. Numeric columns were coerced to float; the species column was kept as a categorical string. No imputation was required.

6. Exploratory Analysis & Visualizations

6.1 Correlation Heatmap

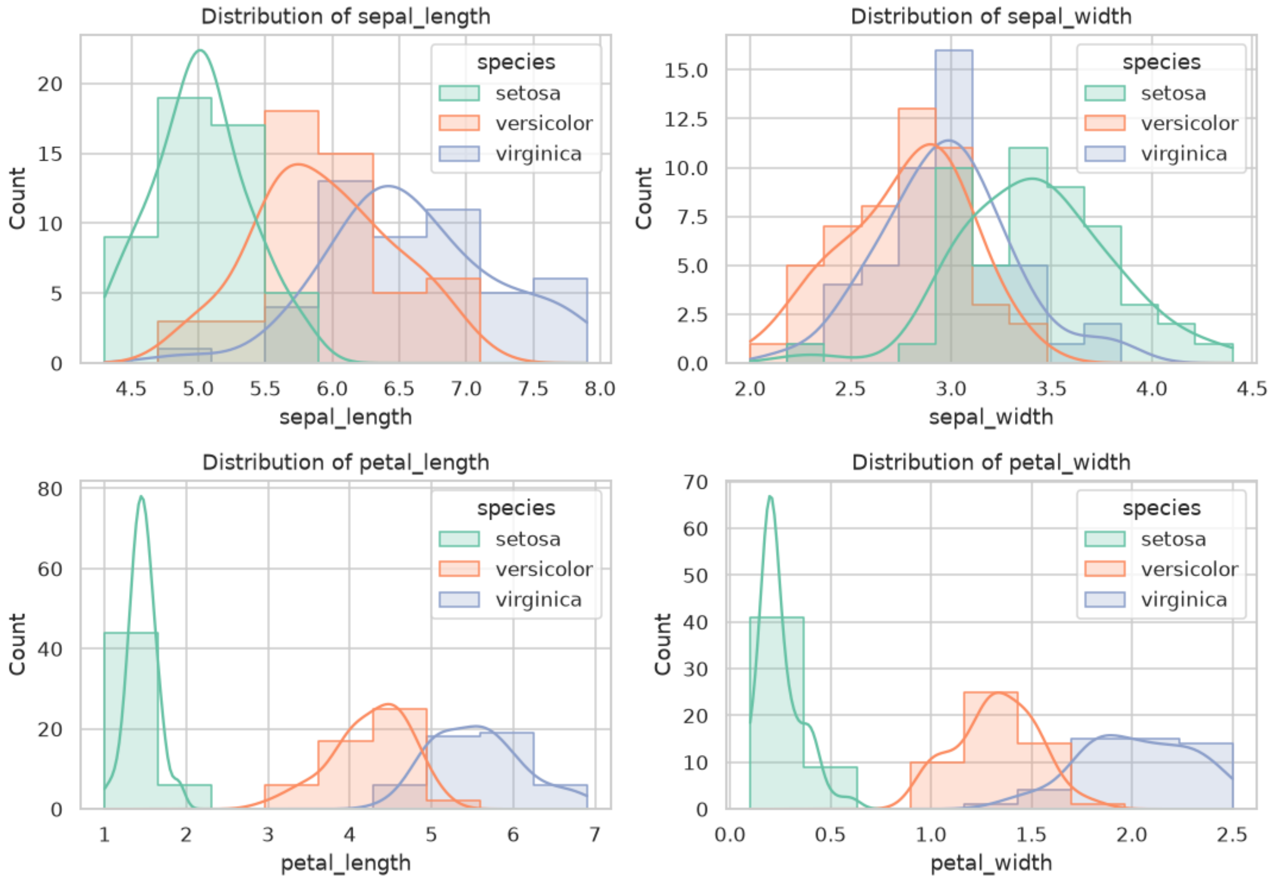
Petal length and petal width are very strongly correlated (~0.96). Sepal width is weakly - and slightly negatively - correlated with the petal dimensions, suggesting it carries somewhat independent information.

6.2 Boxplots by Species



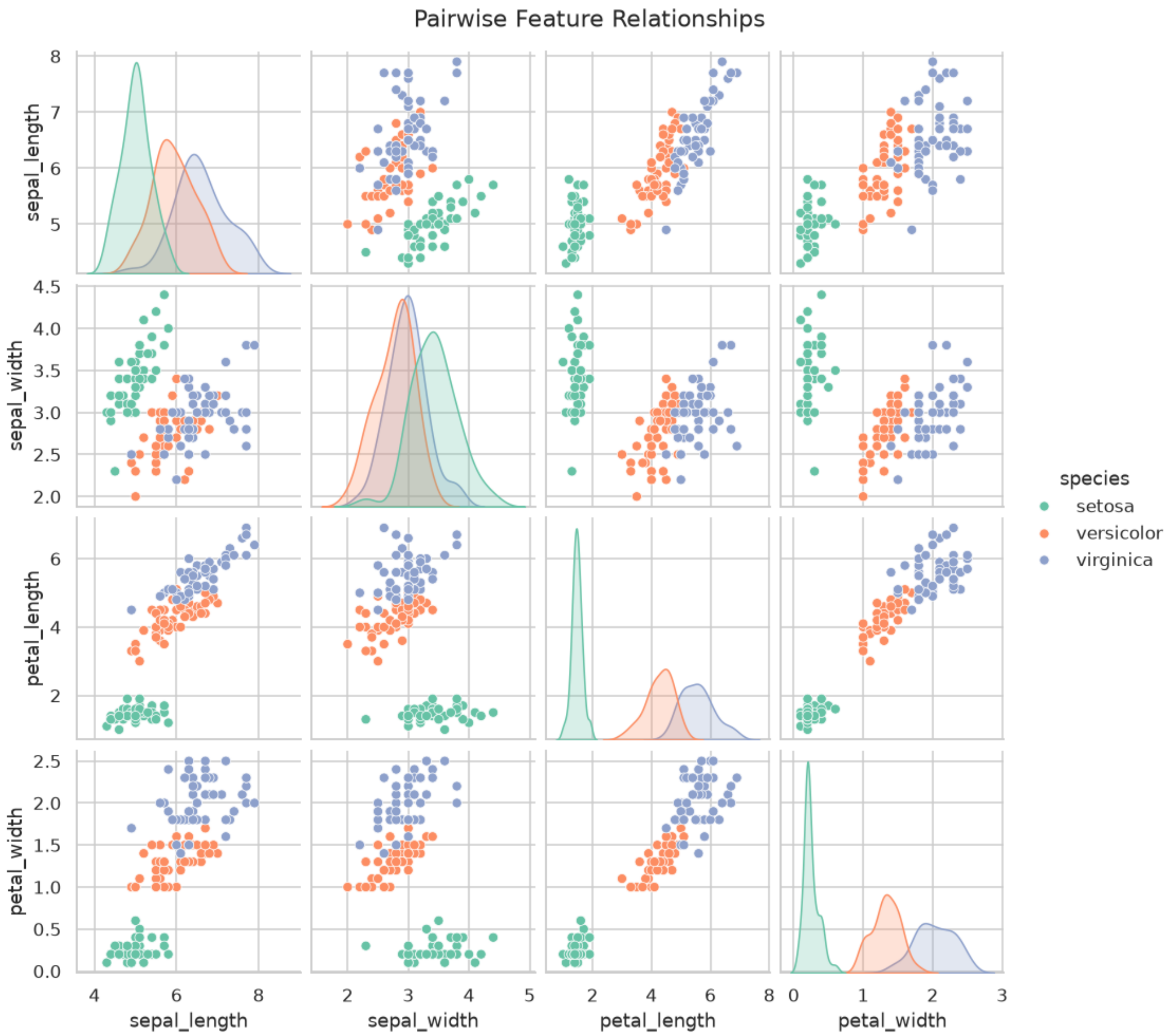
The petal features (length and width) show clear, non-overlapping ranges between setosa and the other two species. A handful of mild outliers appear in sepal_width for setosa but are not extreme.

6.3 Feature Distributions



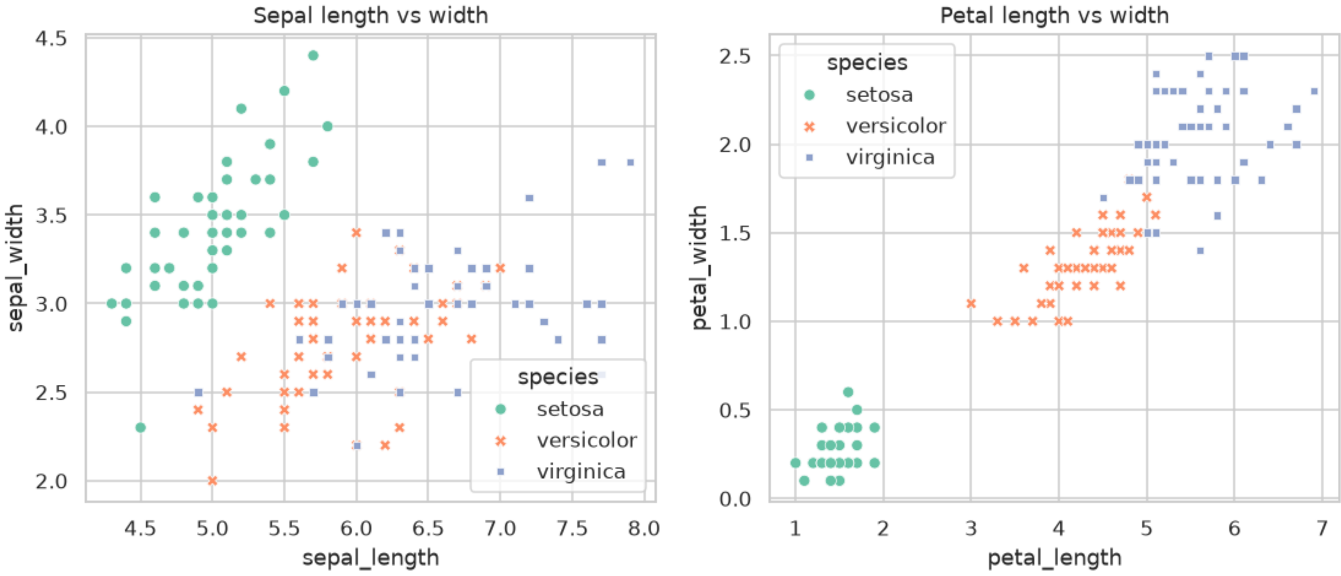
Petal length and width are bimodal across the full dataset, with setosa occupying the lower mode. Sepal length is roughly normal; sepal width is close to normal with slight right skew.

6.4 Pairwise Relationships



The pairplot confirms that setosa is linearly separable from the other two species in any plot involving petal_length or petal_width. Versicolor and virginica overlap slightly but are still well-separated in the petal subspace.

6.5 Sepal vs Petal Scatter



Petal-space scatter reveals three distinct clusters, while sepal-space scatter mixes versicolor and virginica - quantitative confirmation that petal measurements are the dominant discriminators.

7. Findings

- The dataset is clean: no missing values and no duplicates after dedup.
- Petal length & width are the most informative features (corr ~0.96).
- Setosa is linearly separable from versicolor and virginica.
- Versicolor and virginica overlap on sepal features but separate on petals.
- A few mild outliers exist in sepal_width but do not warrant removal.

8. Conclusion

The Iris dataset, despite its small size, is an excellent teaching dataset. EDA alone is sufficient to suggest that a simple classifier using petal length and petal width could achieve very high accuracy. This makes Iris an ideal next step for supervised classification - logistic regression, KNN or decision trees would all be appropriate follow-up models.

9. References

- Fisher, R. A. (1936). 'The use of multiple measurements in taxonomic problems.' Annals of Eugenics, 7(2), 179-188.
- UCI Machine Learning Repository: Iris Data Set.
- scikit-learn documentation: sklearn.datasets.load_iris.
- McKinney, W. (2010). Data Structures for Statistical Computing in Python.
- Waskom, M. (2021). Seaborn: statistical data visualization.